

CHAPTER 10

Cell Cycle and Cell Division

Cell Cycle

1. Match List-I with List-II. (2024 Re)

	List-I		List-II
A.	Cells are metabolically active and proliferate	I.	G ₂ Phase
B.	DNA replication takes place	II.	G ₁ phase
C.	Proteins are synthesised	III.	G ₀ phase
D.	Quiescent stage with metabolically active cells	IV.	S phase

Choose the **correct** answer from the options given below:

- a. A-IV, B-II, C-III, D-I b. A-I, B-III, C-IV, D-II
c. A-II, B-I, C-III, D-IV d. A-II, B-IV, C-I, D-III

2. Following are the stages of cell division : (2024)

- A. Gap 2 phase B. Cytokinesis
C. Synthesis phase D. Karyokinesis
E. Gap 1 phase

Choose the correct sequence of stages from the options given below:

- a. B-D-E-A-C b. E-C-A-D-B
c. C-E-D-A-B d. E-B-D-A-C

3. Among eukaryotes, replication of DNA takes place in (2023)

- a. G₂ phase b. M phase c. S phase d. G₁ phase

4. Given below are two statements: (2023)

Statement-I: During G₀ phase of cell cycle the cell is metabolically inactive.

Statement-II: The centrosome undergoes duplication during S phase of interphase.

In the light of the above statements, choose the most appropriate answer from the option below:

- a. Statement-I is incorrect but Statement-II is correct
b. Both Statement-I and Statement-II are correct
c. Both Statement-I and Statement-II are incorrect.
d. Statement-I is correct but Statement-II is incorrect.

5. Match List-I with List-II (2023)

	List-I		List-II
(A)	M Phase	(I)	Proteins are synthesized
(B)	G ₂ Phase	(II)	Inactive phase
(C)	Quiescent stage	(III)	Interval between mitosis and initiation of DNA replication
(D)	G ₁ Phase	(IV)	Equational division

Choose the correct answer from the options given below:

- a. A-II, B-IV, C-I, D-III b. A-III, B-II, C-IV, D-I
c. A-IV, B-II, C-I, D-III d. A-IV, B-I, C-II, D-III

6. Which one of the following is the quiescent stage of cell cycle? (2023-Manipur)

- a. M b. G₂ c. G₁ d. G₀

7. The fruit fly has 8 chromosomes (2n) in each cell. During interphase of Mitosis if the number of chromosomes at G₁ phase is 8, what would be the number of chromosomes after S phase? (2021)

- a. 16 b. 4 c. 32 d. 8

8. The centriole undergoes duplication during: (2021)

- a. Prophase b. Metaphase
c. G₂ phase d. S-phase

9. Match List-I with List-II (2021)

	List-I		List-II
A.	S phase	(i)	Proteins are synthesized
B.	G ₂ phase	(ii)	Inactive phase
C.	Quiescent stage	(iii)	Interval between mitosis and initiation of DNA replication
D.	G ₁ phase	(iv)	DNA replication

Choose the correct answer from the options given below.

- a. A-iv B-ii C-iii D-i
b. A-iv B-i C-ii D-iii
c. A-ii B-iv C-iii D-i
d. A-iii B-ii C-i D-iv

10. Identify the correct statement with regard to G₁ phase (Gap 1) of interphase. (2020)

- a. Reorganisation of all cell components takes place.
b. Cell is metabolically active, grows but does not replicate its DNA.
c. Nuclear division takes place.
d. DNA synthesis or replication takes place.

11. Some dividing cells exit the cell cycle and enter vegetative inactive stage. This is called quiescent stage (G₀). This process occurs at the end of: (2020)

- a. G₁ phase b. S phase c. G₂ phase d. M phase

12. Match the following events that occur in their respective phases of cell cycle and select the correct option: (2020-Covid)

1.	G ₁ phase	(i)	Cell grows and organelle duplication
2.	S phase	(ii)	DNA replication and chromosome duplication
3.	G ₂ phase	(iii)	Cytoplasmic growth
4.	Metaphase in M-phase	(iv)	Alignment of chromosomes

- a. 1-(iii) 2-(iv) 3-(i) 4-(ii)
 b. 1-(iv) 2-(i) 3-(ii) 4-(iii)
 c. 1-(i) 2-(ii) 3-(iii) 4-(iv)
 d. 1-(ii) 2-(iii) 3-(iv) 4-(i)
13. The correct sequence of phases of cell cycle is (2019)
 a. $M \rightarrow G_1 \rightarrow G_2 \rightarrow S$ b. $G_1 \rightarrow G_2 \rightarrow S \rightarrow M$
 c. $S \rightarrow G_1 \rightarrow G_2 \rightarrow M$ d. $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$
14. Cell in G₀ phase (2019)
 a. Exit the cell cycle b. Enter the cell cycle
 c. Suspend the cell cycle d. Terminate the cell cycle
15. DNA replication in bacteria occurs: (2017-Delhi)
 a. During S-phase b. Within nucleolus
 c. Prior to fission d. Just before transcription
16. When cell has stalled DNA replication fork, which checkpoint should be predominantly activated? (2016 - II)
 a. M b. Both G₂/M and M
 c. G₁/S d. G₂/M
17. During cell growth, DNA synthesis takes place in: (2016 - II)
 a. G₂ phase b. M phase c. S phase d. G₁ phase
18. A somatic cell that has just completed the S phase of its cell cycle, as compared to gamete of the same species, has: (2015 Re)
 a. Twice the number of chromosomes and four times the amount of DNA
 b. Four times the number of chromosomes and twice the amount of DNA
 c. Twice the number of chromosomes and twice the amount of DNA
 d. Same number of chromosomes but twice the amount of DNA
19. During which phase(s) of cell cycle, amount of DNA in a cell remains at 4C level if the initial amount is denoted as 2C? (2014)
 a. G₂ and M b. G₀ and G₁
 c. G₁ and S d. Only G₂
20. In 'S' phase of the cell cycle: (2014)
 a. Amount of DNA is reduced to half in each cell
 b. Amount of DNA doubles in each cell
 c. Amount of DNA remains same in each cell
 d. Chromosome number is increased

Mitosis

21. Spindle fibers attach to kinetochores of chromosomes during (2024)
 a. Anaphase b. Telophase
 c. Prophase d. Metaphase
22. Doubling of the number of chromosomes can be achieved by disrupting mitotic cell division soon after: (2023-Manipur)
 a. Anaphase b. Telophase
 c. Prophase d. Metaphase
23. With respect to metaphase, which of the following statements is incorrect? (2022 Re)
 a. Chromosomes lie at the equator of the cell
 b. Complete disintegration of nuclear envelope takes place
 c. Chromosomes are highly condensed
 d. Metaphase chromosomes are made up of four sister chromatids held together by centromere
24. Which one of the following never occurs during mitotic cell division? (2022)
 a. Coiling and condensation of the chromatids
 b. Spindle fibres attach to kinetochores of chromosomes
 c. Movement of centrioles towards opposite poles
 d. Pairing of homologous chromosomes
25. Select the incorrect statement with reference to mitosis: (2022)
 a. Splitting of centromere occurs at anaphase.
 b. All the chromosomes lie at the equator at metaphase.
 c. Spindle fibres attach to centromere of chromosomes.
 d. Chromosomes decondense at telophase.
26. In a mitotic cycle, the correct sequence of phases is (2020-Covid)
 a. G₁, S, G₂, M b. M, G₁, G₂, S
 c. G₁, G₂, S, M d. S, G₁, G₂, M
27. Attachment of spindle fibers to kinetochores of chromosomes becomes evident in : (2020-Covid)
 a. Telophase b. Prophase c. Metaphase d. Anaphase
28. Which of the following options gives the correct sequence of events during mitosis? (2017-Delhi)
 a. Condensation → Nuclear membrane disassembly → Crossing over → Segregation → Telophase
 b. Condensation → Nuclear membrane disassembly → Arrangement at equator → Centromere division → Segregation → Telophase
 c. Condensation → Crossing over → Nuclear membrane disassembly → Segregation → Telophase
 d. Condensation → Arrangement at equator → Centromere division → Segregation → Telophase
29. Anaphase promoting complex (APC) is a protein degradation machinery necessary for proper mitosis of animal cells. If APC is defective in a human cell, which of the following is expected to occur? (2017-Delhi)
 a. Chromosomes will not condense
 b. Chromosomes will be fragmented
 c. Chromosomes will not segregate
 d. Recombination of chromosome arms will occur
30. Which of the following is not a characteristic feature during mitosis in somatic cells? (2016 - I)
 a. Spindle fibres b. Disappearance of nucleolus
 c. Chromosome movement d. Synapsis

31. Spindle fibres attach on to: (2016 - I)
 a. Telomere of the chromosome
 b. Kinetochore of the chromosome
 c. Centromere of the chromosome
 d. Kinetosome of the chromosome
32. A cell at telophase stage is observed by a student in a plant brought from the field. He tells his teacher that this cell is not like other cells at telophase stage. There is no formation of cell plate and thus the cell is containing more number of chromosomes as compared to other dividing cells. This would result in: (2016 - I)
 a. Aneuploidy b. Polyploidy
 c. Somaclonal variation d. Polyteny
33. A stage in cell division is shown in the figure. Select the answer which gives correct identification of the stage with its characteristics: (2013)



a.	Telophase	Endoplasmic reticulum and nucleolus not reformed yet.
b.	Telophase	Nuclear envelope reforms, Golgi complex reforms.
c.	Late Anaphase	Chromosomes move away from equatorial plate, Golgi complex not present.
d.	Cytokinesis	Cell plate formed, mitochondria distributed between two daughter cells.

Meiosis

34. Given below are two statements: (2024)
Statement-I: Chromosomes become gradually visible under light microscope during leptotene stage.
Statement-II: The beginning of diplotene stage is recognized by dissolution of synaptonemal complex.
 In the light of the above statements, choose the correct answer from the options given below:
 a. Statement-I is true but Statement-II is false
 b. Statement-I is false but Statement-II is true
 c. Both Statement-I and Statement-II are true
 d. Both Statement-I and Statement-II are false
35. Match List-I with List-II: (2024)

List-I (Sub Phases of Prophase I)		List-II (Specific characters)	
(A)	Diakinesis	(I)	Synaptonemal complex formation
(B)	Pachytene	(II)	Completion of terminalisation of chiasmata
(C)	Zygotene	(III)	Chromosomes look like thin threads
(D)	Leptotene	(IV)	Appearance of recombination nodules

Choose the correct answer from the options given below;

- a. A-(II), B-(IV), C-(I), D-(III)
 b. A-(IV), B-(III), C-(II), D-(I)
 c. A-(IV), B-(II), C-(III), D-(I)
 d. A-(I), B-(II), C-(IV), D-(III)
36. Recombination between homologous chromosomes is completed by the end of: (2024 Re)
 a. Diakinesis b. Zygotene
 c. Diplotene d. Pachytene
37. Match List-I with List-II. (2024 Re)

List-I Event		List-II Stage of Prophase-I (Meiosis-I)	
(A)	Chiasmata formation	(I)	Pachytene
(B)	Crossing over	(II)	Diakinesis
(C)	Synaptonemal complex formation	(III)	Diplotene
(D)	Terminalisation of chiasmata	(IV)	Zygotene

Choose the **correct** answer from the options given below:

- a. A-III, B-I, C-IV, D-II b. A-II, B-I, C-III, D-IV
 c. A-III, B-I, C-II, D-IV d. A-II, B-III, C-IV, D-I
38. Which of the following stages of meiosis involves division of centromere? (2023)
 a. Telophase b. Metaphase I
 c. Metaphase II d. Anaphase II
39. The process of appearance of recombination nodules occurs at which sub stage of prophase I in meiosis? (2023)
 a. Diakinesis b. Pachytene
 c. Zygotene d. Diplotene
40. Select the correct statements. (2023)
 A. Tetrad formation is seen during leptotene.
 B. During Anaphase, the centromere split and chromatids separate.
 C. Terminalization takes place during Pachytene.
 D. Nucleolus, Golgi complex and ER are reformed during Telophase.
 E. Crossing over takes place between sister chromatids of homologous chromosome
 Choose the correct answer from the option given below:
 a. B and E only b. A and C only
 c. B and D only d. A, C and E only
41. The dissolution of synaptonemal complex occurs during: (2023-Manipur)
 a. Pachytene b. Diplotene
 c. Diakinesis d. Leptotene
42. During which stages of mitosis and meiosis, respectively does the centromere of each chromosome split? (2023-Manipur)
 a. Metaphase, Metaphase II
 b. Prophase, Telophase I
 c. Telophase, Anaphase I
 d. Anaphase, Anaphase II

43. Select **correct** sequence of substages of Prophase-I of Meiotic division: (2023-Manipur)

(A) Zygotene (B) Pachytene (C) Diakinesis
(D) Leptotene (E) Diplotene

Choose the **correct** answer from the options given below:

a. (D), (B), (A), (E), (C) b. (A), (B), (D), (E), (C)
c. (D), (A), (B), (E), (C) d. (A), (D), (B), (C), (E)

44. Which stage of meiosis can last for months or years in the oocytes of some vertebrates? (2022 Re)

a. Diakinesis b. Leptotene
c. Pachytene d. Diplotene

45. In meiosis, crossing over and exchange of material between homologous chromosomes catalyzed by the enzyme: (2022 Re)

a. Polymerase b. Phosphorylase
c. Recombinase d. Transferase

46. Identify the correct sequence of events during Prophase I of meiosis: (2022 Re)

A. Synapsis of homologous chromosomes
B. Chromosomes become gradually visible under microscope
C. Crossing over between non-sister chromatids of homologous chromosomes
D. Terminalisation of chiasmata
E. Dissolution of synaptonemal complex

Choose the correct answer from the options given below:

a. (A), (C), (D), (E), (B) b. (A), (B), (C), (D), (E)
c. (B), (C), (D), (E), (A) d. (B), (A), (C), (E), (D)

47. Bivalent or Tetrad formation is a characteristic feature observed during: (2022 Re)

a. Chiasmata in zygotene stage
b. Synaptonemal complex in zygotene stage
c. Chiasmata in Diplotene stage
d. Synaptonemal complex in Pachytene stage

48. The appearance of recombination nodules on homologous chromosomes during meiosis characterizes (2022)

a. Terminalization
b. Synaptonemal complex
c. Bivalent
d. Sites at which crossing over occurs

49. Regarding Meiosis, which of the statements is incorrect? (2022)

a. Four haploid cells are formed at the end of Meiosis-II
b. There are two stages in Meiosis, Meiosis-I and II
c. DNA replication occurs in S phase of Meiosis-II
d. Pairing of homologous chromosomes and recombination occurs in Meiosis-I

50. Which stage of meiotic prophase shows terminalisation of chiasmata as its distinctive feature? (2021)

a. Zygotene b. Diakinesis
c. Pachytene d. Leptotene

51. Which of the following stages of meiosis involves division of centromere? (2021)

a. Metaphase II b. Anaphase II
c. Telophase II d. Metaphase I

52. Match the following with respect to meiosis: (2020)

	Column-I		Column-II
1.	Zygotene	(i)	Terminalization
2.	Pachytene	(ii)	Chiasmata
3.	Diplotene	(iii)	Crossing over
4.	Diakinesis	(iv)	Synapsis

Select the correct option from the following:

(1) (2) (3) (4) (1) (2) (3) (4)

a. (iv)(iii)(ii)(i) b. (i)(ii)(iv)(iii)
c. (ii)(iv)(iii)(i) d. (iii)(iv)(i)(ii)

53. Dissolution of the synaptonemal complex occurs during: (2020)

a. Zygotene b. Diplotene
c. Leptotene d. Pachytene

54. During Meiosis I, in which stage synapsis takes place? (2020-Covid)

a. Zygotene b. Diplotene
c. Leptotene d. Pachytene

55. The stage during which separation of the paired homologous chromosomes begins is (2018)

a. Pachytene b. Diplotene
c. Diakinesis d. Zygotene

56. Match the stages of meiosis in Column-I to their characteristic features in Column-II and select the correct option using the codes given below: (2016 - II)

	Column-I		Column-II
A.	Pachytene	i.	Pairing of homologous chromosomes
B.	Metaphase-I	ii.	Terminalisation of chiasmata
C.	Diakinesis	iii.	Crossing over takes place
D.	Zygotene	iv.	Chromosomes align at equatorial plate

Codes:

a. A-ii B-iv C-iii D-i
b. A-iv B-iii C-ii D-i
c. A-iii B-iv C-ii D-i
d. A-i B-iv C-ii D-iii

57. In meiosis, crossing over is initiated at: (2016 - I)

a. Pachytene b. Leptotene
c. Zygotene d. Diplotene

58. Select the correct option: (2015)

	Column-I		Column-II
A.	Synapsis aligns the homologous chromosomes	i.	Anaphase-II
B.	Synthesis of RNA and protein	ii.	Zygotene
C.	Action of enzyme recombinase	iii.	G ₂ -phase
D.	Centromeres do not separate but chromatids move towards opposite poles	iv.	Anaphase-I
		v.	Pachytene

